

Duke Essays Tony

Intellectual experience

Tell us about an intellectual experience in the past two years that you found absolutely fascinating.

(250 word limit)

In late 2023, I was invited to a national math olympiad camp to receive training ahead of the 2024 international Olympiads. My lecturer, Dr. James, mentioned a fascinating algorithm that I was already familiar with: The Euclidean Algorithm. It was used to quickly find the greatest common divisor (GCD) of a pair of numbers.

However, a random idea popped into my mind as I left the class. I noticed a car driving forward, then stopping and reversing. That was when it hit me: Is it possible to reverse the Euclidean Algorithm? Given the GCD, could I find the two numbers? This would be a revolutionary method used to solve many number theoretical problems.

At that moment, I was buoyed with eclectic joy. Dashing to get a book and a pen, I scribbled some numbers and statements before asking myself again: Is it possible? My joy clouded my thoughts, hence I took several deep breaths and got to work afresh. Manipulating numbers and equations, I waded through the sea of chaos and successfully ended up at the answer. Indeed, I rushed to tell my lecturer about my latest discovery.

To my utter dismay, he revealed that the method I employed was standard in solving linear Diophantine equations. I was annoyed that mathematicians got there centuries before me, but I'll always remember the moment of joyous discovery. Seeking the unknown with curiosity will forever be a captivating cherished pastime, whether or not I get there first.

Why Duke?

What is your sense of Duke as a university and a community, and why do you consider it a good match for you? If there's something in particular about our offerings that attracts you, feel free to share that as well.

(250 word limit)

Duke University appeals to me because of its dedication to promoting diversity and inclusiveness, not to mention its cutting-edge academic programs and lively intellectual community. As a black student who is passionate about solving mathematical puzzles, Duke's commitment to offering unmatched possibilities perfectly fits my objectives.

Conversations with leading academics such as Professor Ingrid Daubechies at Duke's Department of Mathematics will broaden my lens and give me new perspectives in the mathematical landscape. The tightly-knit community in the department deeply resonates with me, enabling me to find a place among passionate mathematicians from all over the world.

Duke's emphasis on interdisciplinary studies—exemplified by initiatives like the Bass Connections program—perfectly matches my goal of fusing mathematics with other disciplines. I am excited to investigate how mathematics may be used practically in interdisciplinary teams to solve real-world problems. At the Network Analysis Center, I will be thrilled to use my mathematical expertise to study complex issues such as disease spread.

Beyond the classroom, Duke's dedication to community involvement, demonstrated by distinct initiatives like DukeEngage, is in line with my goal of using problem-solving abilities to address global societal issues. Giving back to communities around me will certainly be a cornerstone of the Duke experience, fostering in me greater empathy.

These three words remind me of Duke: Curiosity, Diversity, and Service. Truly, it is the place where I would simultaneously enrich myself and those around me.